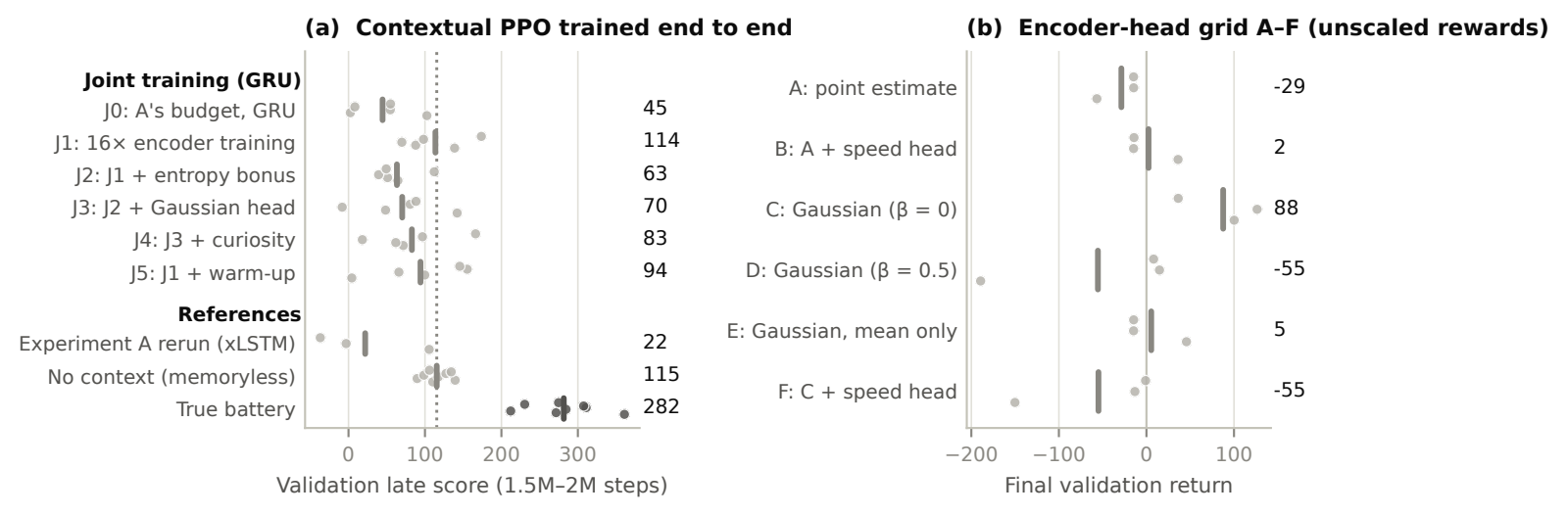
(a) Contextual PPO trained end to end
Joint training (GRU)
J0: A's budget, GRU
J1: 16× encoder training
J2: J1 + entropy bonus
J3: J2 + Gaussian head
J4: J3 + curiosity
J5: J1 + warm-up
References
Experiment A rerun (xLSTM)
No context (memoryless)
True battery
45
114
63
70
83
94
22
115
282
0
100
200
300
Validation late score (1.5M–2M steps)
(b) Encoder-head grid A–F (unscaled rewards)
A: point estimate
B: A + speed head
C: Gaussian (β = 0)
D: Gaussian (β = 0.5)
E: Gaussian, mean only
F: C + speed head
-29
2
88
-55
5
-55
−200
−100
0
100
Final validation return